

Bacterial preQ1 Incorporation and Queuosine Maturation

A concise, review-style synthesis for a molecular-biology audience Search date: 2026-07-27. Databases: PubMed. This review is anchored to primary literature and authoritative reviews; PMIDs are given for major claims.

1. Executive summary

Queuosine (Q) is a 7-deazaguanosine hypermodification found at the wobble position (nucleotide 34) of the four bacterial and eukaryotic tRNAs that read GUN codons — tRNA(Asp), tRNA(Asn), tRNA(His) and tRNA(Tyr). This review treats the **bacterial four-enzyme module that converts the precursor preQ0 into mature Q on tRNA**: (1) QueF reduces the nitrile of preQ0 to the primary amine preQ1; (2) Tgt (tRNA-guanine transglycosylase) exchanges guanine-34 for preQ1 by base exchange; (3) QueA transfers and isomerizes the ribosyl moiety of S-adenosylmethionine (SAM) onto preQ1-tRNA to form epoxyqueuosine (oQ); and (4) an epoxyqueuosine reductase (QueG or, in many lineages, QueH) reduces/deoxygenates oQ to the cyclopentene ring of mature Q

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The module is strictly ordered and largely obligatory once the tRNA is committed at the Tgt step. Its defining features are chemically unusual: QueF performs the only known biological nitrile-to-amine reduction

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QueA uses SAM in a non-methyl-transfer, ribosyl-transfer role
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and the final step is solved by two non-homologous enzymes — cobalamin/[4Fe-4S]-dependent QueG (a Class III cobalamin enzyme related to reductive dehalogenases) or cobalamin-independent QueH

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The complete de novo route exists **only in bacteria**; eukaryotes are queuine auxotrophs that salvage the free base and insert it with a heterodimeric TGT
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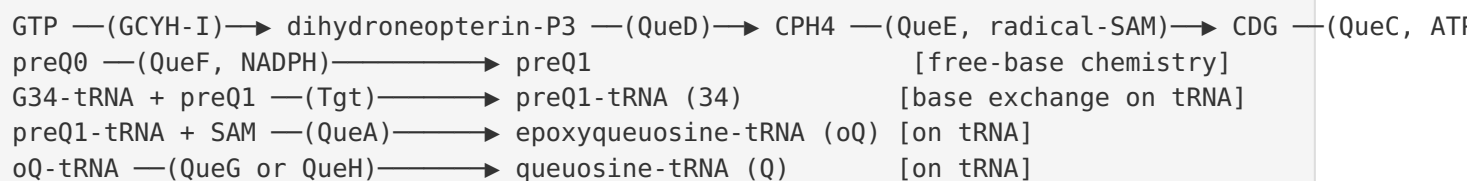
The upstream precursor preQ0 (made from GTP by QueC/QueD/QueE) is shared with a distinct **DNA 7-deazaguanine modification** system used in phage/host defense — a neighboring pathway frequently confused with tRNA Q biosynthesis but mechanistically separate
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2. Definition and biological boundaries

2.1 What is included

The system, as scoped here, is the **post-transcriptional maturation of Q on tRNA in bacteria**, comprising four catalytic steps acting on a common linear intermediate flow:



Steps 1–2 up to preQ1 formation are free-base metabolism; steps 3–5 (Tgt insertion, QueA, QueG/H) occur on the intact tRNA

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The four substrate tRNAs (Asp, Asn, His, Tyr) all carry a GUN anticodon with a UGU recognition motif read by Tgt

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2.2 Neighboring processes that should be treated separately

- **preQ0 de novo biosynthesis (QueC/QueD/QueE from GTP).** Often bundled with "Q biosynthesis," but chemically upstream and separable; a *Streptomyces* cluster bearing only queC/queD/queE (no queF) accumulates preQ0 as an end-product with no tRNA modification

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The provisional brief begins the module at "preQ1 formation," so QueC/D/E are boundary/context rather than core.

- **DNA 7-deazaguanine modification (DpdA/tgtA5 systems).** A paralogous transglycosylase (DpdA) inserts 7-deazaguanine derivatives (dPreQ0, dADG, and phage-specific mdPreQ1/fdPreQ1) into DNA for restriction-modification/anti-CRISPR defense in phages and diverse bacteria. It shares the preQ0 precursor and a TGT-fold enzyme but is a different macromolecular target and biological function

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- **Archaeosine (G+) biosynthesis in archaea.** Archaeal TGT (ArcTGT) exchanges guanine for preQ0 in the **D-loop (position 15)** of tRNA, not the anticodon, and the downstream chemistry differs entirely (no QueA/QueG). It is a sister branch of the TGT superfamily, not

part of the bacterial Q anticodon pathway

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- **Eukaryotic queuine salvage.** Eukaryotes do not synthesize Q; they salvage dietary/microbiota-derived queuine and insert it with a heterodimeric TGT (QTRT1/QTRT2), then require a separate 2'-O-methylation on tRNA(Asp) in some taxa. This is a salvage branch sharing only the base-exchange concept

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2.3 Competing definitions

Literature varies on where the "Q pathway" starts. Some reviews include GTP→preQ0 (QueC/D/E) as step 1; others (as in this brief) start at preQ1 formation. There is also terminological drift between "QueF" as a specific enzyme and "the nitrile reductase family," and between "queuosine biosynthesis" (bacterial, tRNA) and "deazaguanine modification" (which now spans tRNA and DNA). Being explicit about the target macromolecule (tRNA vs DNA) and the start metabolite (GTP vs preQ0 vs preQ1) resolves most ambiguity.

3. Mechanistic overview

3.1 Best current model of the sequence of events

1. **preQ1 formation (QueF).** QueF catalyzes the NADPH-dependent 4-electron reduction of the nitrile of preQ0 to the primary amine preQ1 — the only nitrile-to-amine reduction known in biology

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Catalysis proceeds through a covalent thioimide intermediate on a conserved active-site cysteine (Cys55 in *Bacillus subtilis*; Cys190 in *E. coli*), then a second hydride transfer to the amine; an imine intermediate is sequestered during turnover

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Substrate binding, not thioimide formation, triggers active-site closure

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2. **preQ1 insertion (Tgt).** Bacterial TGT performs an **irreversible base-exchange** by a **double-displacement mechanism** through a covalent enzyme–RNA intermediate

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A conserved active-site aspartate is the nucleophile that attacks C1' of ribose-34 after guanine departs (Asp102 in *Zymomonas mobilis*; its Ala mutant is inactive and cannot form the covalent intermediate —

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a second conserved aspartate (Asp264 in *E. coli*) mediates the proton transfers required to protonate the leaving guanine and deprotonate the incoming base (only the conservative D264E mutant retains activity —

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and a distinct aspartate (Asp156 in *Z. mobilis*) is the principal preQ1-recognition element in the substrate pocket

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The anticodon stem-loop (ASL) bearing the UGU motif is the minimal substrate; full-length tRNA binds with higher affinity through additional contacts with the dimeric enzyme

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This is the committed step that places the precursor on the tRNA.

3. **Epoxyqueuosine formation (QueA).** QueA (SAM:tRNA ribosyltransferase-isomerase) forms oQ from preQ1-tRNA in a **single SAM-requiring step**, transferring and isomerizing the ribosyl moiety of SAM (not its methyl group) to build the 2,3-epoxy-4,5-dihydroxycyclopentane ring appended to the preQ1 aminomethyl group

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A 17-nt anticodon minihelix suffices as the RNA substrate; measured parameters include

Km(SAM) $\approx$ 101 μ M, Km(tRNA) $\approx$ 1.5 μ M, kcat $\approx$ 2.5 min⁻¹, optimal pH $\approx$ 8.7, with mM Mg²⁺/Mn²⁺ inhibitory
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4. **Epoxyqueuosine reduction (QueG or QueH).** The final reduction/deoxygenation converts the epoxide to the cyclopentene, yielding mature Q
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QueG is cobalamin- and twin-[4Fe-4S]-dependent, structurally a Class III cobalamin enzyme; it carries a redox chain of two [4Fe-4S] clusters plus base-off cob(II)alamin and is proposed to act via Co–C bond formation with conserved Tyr/Asp acting as proton donors to the epoxide

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Many organisms that lack QueG use **QueH**, a cobalamin-independent, structurally distinct metalloenzyme catalyzing the same step
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3.2 Obligatory, conditional, and accessory steps

- **Obligatory (core):** QueF→Tgt→QueA→(QueG or QueH). Each product is the substrate of the next; skipping any step blocks maturation and leaves a defined intermediate on tRNA (preQ1-tRNA in queA mutants; oQ-tRNA in queG/queH mutants)
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- **Conditional / lineage-specific:** The identity of the final reductase (QueG vs QueH) is a non-orthologous alternative — one or the other, depending on lineage
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- **Accessory / bypass:** preQ0/preQ1/queuine **salvage and transport** (e.g., the YhhQ preQ0 transporter and novel salvage enzymes in pathogens) can supply the precursor and bypass early de novo steps, so Tgt activity does not imply full de novo capacity

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4. Major molecular players and active assemblies

Step	Enzyme	Reaction	Cofactor / chemistry	Assembly	Key refs
1	QueF	preQ0 → preQ1	NADPH; covalent thioimide on catalytic Cys; only biological nitrile reductase	Tunnel-fold (T-fold) homodecamer (2 head-to-head pentamers), 10 interfacial active sites; "unimodular" and "bimodular" QueF subtypes exist	22787148, 28300774, 29339556
2	Tgt	G34 + preQ1 → preQ1-tRNA ³⁴ + G	Double-displacement base exchange via covalent Asp102-ribose intermediate (Asp264 proton transfer; Asp156 preQ1 recognition); no cofactor	Homodimer (bacteria); (β/α)8 TIM-barrel + zinc-binding subdomain; ASL/UGU recognition	39956694, 8961936, 12909636, 21131277
3	QueA	preQ1-tRNA + SAM → oQ-tRNA	SAM ribosyl transfer + isomerization (non-methyl role)	Monomeric two-domain enzyme; acts on ASL minihelix	8347586, 12533518
4	QueG	oQ-tRNA → Q-tRNA	Cobalamin (base-off cob(II)alamin) + two [4Fe-4S]; Co-C bond mechanism; Class III cobalamin enzyme (RdhA-like)	Monomer with redox chain; docks tRNA anticodon	27638883, 26378237
4'	QueH	oQ-tRNA → Q-tRNA	Cobalamin- independent unique metalloenzyme	Structurally distinct from QueG	34652139

Step	Enzyme	Reaction	Cofactor / chemistry	Assembly	Key refs
upstream	GCYH-I + QueD + QueE + QueC	GTP → CDG → preQ0	GTP cyclohydrolase I; QueD = 6-carboxy-5,6,7,8-tetrahydropterin (CPH4) synthase; QueE = radical-SAM CDG synthase (ring rearrangement); QueC = ATP-dependent nitrile synthase (CDG→preQ0)	multi-enzyme	25353335, 30097106, 28045519
accessory	YhhQ / salvage enzymes	preQ0/ queuine import & reuse	membrane transporter (COG1738); salvage kinases/glycosylases	—	28208705, 31481610
eukaryotic salvage	QTRT1/ QTRT2 (eukaryotic TGT)	G34 + queuine → Q-tRNA ³⁴	base exchange; inserts the free base queuine, not preQ1	obligate heterodimer	39956693, 35815944, 21131277

Substrate-specificity logic in TGT. Divergent evolution tuned the TGT active site to its cognate 7-deazaguanine: bacterial TGT uses a Cys (e.g., Cys145 in *E. coli*) to recognize preQ1, whereas the eukaryotic enzyme substitutes a Val to favor queuine over preQ1 (

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This single-residue logic is why bacteria insert preQ1 (then mature it in situ) while eukaryotes insert the finished base.

5. Evolutionary and cell-biological variation

5.1 Across evolutionary lineages

- **Bacteria:** full de novo pathway (QueC/D/E→QueF→Tgt→QueA→QueG/H). Final-step reductase is **QueG or QueH** depending on lineage — a textbook non-orthologous gene

displacement

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- **Eukaryotes:** no de novo synthesis; queuine auxotrophy with salvage and insertion by heterodimeric QTRT1/QTRT2. In many eukaryotes tRNA(Asp) Q is further modified by 2'-O-mannosyl/galactosyl or ribose methylation, elaborations absent in bacteria

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- **Archaea:** ArcTGT makes archaeosine at D-loop position 15 using preQ0 directly; no anticodon Q pathway (a sister TGT branch)

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- **Phages/mobile elements:** DpdA/tgtA5 paralogs redeploy the deazaguanine transglycosylase chemistry onto **DNA**, inserting preQ0/preQ1-derived bases for defense; these are horizontally transferred and phylogenetically widespread

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5.2 Origin and ancestral representatives

- **Deep origin:** The 7-deazaguanine/deazapurine metabolism is ancient and modular, spanning tRNA (Q, archaeosine) and DNA modification across all domains and their viruses

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The TGT fold and the T-fold (QueF/GCYH-I family) are old scaffolds.

- **QueF family:** QueF is evolutionarily derived from the GTP cyclohydrolase I / tunnel-fold (T-fold) superfamily; "unimodular" QueF (single T-fold domain, decameric) versus "bimodular" QueF (two fused T-fold domains, pentameric) subtypes exist, with the unimodular form generally regarded as closer to the ancestral GTP-cyclohydrolase-like architecture

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- **Best ancestral representatives:** For understanding the ancestral base-exchange role of TGT, the **bacterial homodimeric Tgt** (e.g., *Zymomonas mobilis*/*E. coli*) is the best-characterized and most representative; the eukaryotic heterodimer and archaeal ArcTGT are derived elaborations with altered substrate/target

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For the final step, **QueG** was the first characterized and provides the mechanistic template (cobalamin/[4Fe-4S]), while QueH represents an independent, later-recognized solution

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5.3 Physiological/state variation

Q levels vary with growth phase, precursor (queueine/preQ) availability, and stress; Q at the wobble position tunes decoding speed and fidelity of GUN codons and buffers translation under stress. In eukaryotes, dependence on gut-microbiota-supplied queueine links host translation to the microbiome, and Q loss affects oxidative-stress response, mitochondrial function, and protein folding

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These functional/physiological data derive largely from eukaryotic systems and should not be assumed identical in bacteria.

6. Constraints, dependencies, and failure modes

- **Strict step order.** Because each enzyme's substrate is the previous product, the sequence QueF→Tgt→QueA→QueG/H is obligatory. QueA cannot act before Tgt has placed preQ1 on tRNA; QueG/H cannot act before QueA has made oQ

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- **Commitment and irreversibility at the insertion step.** The Tgt reaction proceeds through a covalent Asp102–ribose intermediate (double displacement); because guanine is physically released and replaced, the base-exchange is effectively unidirectional *in vivo*, committing that tRNA to maturation

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- **On-tRNA constraint.** Only step 1 (QueF) is free-base chemistry. Tgt, QueA, and QueG/H all require the tRNA (or an ASL minihelix) as substrate; this rules out a purely free-base route to mature Q in which Q is synthesized off-tRNA and inserted — bacteria insert preQ1 and mature it *in situ*

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(Eukaryotes are the exception: they insert the pre-made base queuine, which is why eukaryotic salvage cannot be extrapolated to bacterial mechanism.)

- **Mutually exclusive final enzymes.** QueG and QueH are alternative solutions to the same reaction; an organism's route is determined by which gene it carries

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- **Cofactor dependencies as failure points.** QueF's catalytic Cys is oxidation-prone and is protected *in vivo* by a conserved intramolecular disulfide; loss of this protection inactivates the enzyme

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QueG requires cobalamin and assembled [4Fe-4S] clusters — B12 limitation or defective Fe-S biogenesis can block Q maturation, whereas QueH-using organisms are B12-independent

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- **Defined stall intermediates.** Pathway lesions leave diagnostic intermediates on tRNA: preQ1-tRNA accumulates in queA mutants; oQ-tRNA accumulates in queG/queH mutants

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These are the experimental fingerprints used to order the pathway.

- **Substrate specificity.** TGT selects its 7-deazaguanine by active-site residues (Cys vs Val), so mis-specification (e.g., inserting queuine vs preQ1) is prevented at the insertion step

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7. Controversies and open questions

Strongly supported: - The four-step logic and the identity/order of QueF, Tgt, QueA, and the epoxyqueuosine reductase are established by enzymology, genetics, and structures

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- QueF's unique nitrile-reduction chemistry and thioimide intermediate

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- QueA's single-step SAM ribosyl-transfer/isomerization

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- QueG's cobalamin/[4Fe-4S] redox chain and RdhA relationship

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Debated / indirect / organism-mixing: - Final-step mechanism details. Whether QueG proceeds strictly via Co–C bond formation with Tyr/Asp proton donors is a structure-based proposal that remains to be fully validated kinetically (

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QueH's mechanism and metal identity are newly described and not yet resolved to the depth of QueG

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- **Extrapolation across domains.** Much functional data (decoding, stress, disease links) come from eukaryotes, where the pathway is only a salvage branch; applying these phenotypes to bacterial Q biology risks conflating distinct systems

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- **QueF architecture and ancestry.** The functional and evolutionary significance of unimodular vs bimodular QueF, and the exact path from GTP cyclohydrolase I to QueF, is inferred from structure/phylogeny rather than ancestral reconstruction

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- **Scope of the deazaguanine "arms race."** The full diversity of DNA deazaguanine modifications and their enzymes is still expanding (four new derivatives reported in 2023), so the boundary between "Q pathway" genes and "DNA-defense" paralogs continues to shift

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Most important open questions: 1. What is the definitive catalytic mechanism and metal cofactor of QueH, and why did two unrelated enzymes converge on oQ reduction? 2. How is pathway flux regulated in bacteria (precursor sensing, coupling of de novo vs salvage), and how does Q status feed back on translation under stress? 3. What determines tRNA selectivity and the order in which the four isoacceptors are modified in vivo? 4. How ancient is the split between tRNA-Q and DNA-deazaguanine functions, and which extant enzyme best represents the ancestral transglycosylase?

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